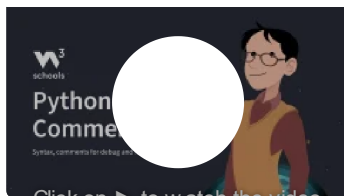


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HTML Canvas Reference

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The **<canvas>** element defines a **bitmapped** area in an HTML page.

The **Canvas API** allows JavaScript to **draw graphics** on the canvas.

The Canvas API can draw shapes, lines, curves, boxes, text, and images, with colors, rotations, transparencies, and other pixel manipulations.

You can add a canvas element anywhere in an HTML page with the **<canvas>** tag:

Example

```
<canvas id="myCanvas" width="300" height="150"></canvas>
```

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```
const myCanvas = document.getElementById("myCanvas");  
const ctx = myCanvas.getContext("2d");
```



Note

The HTML `<canvas>` element itself has no drawing abilities.

You must use JavaScript to draw any graphics.

The `getContext()` method returns an object with tools (methods) for drawing.

Drawing on the Canvas

After you have created a 2D context, you can draw on the canvas.

The **fillRect()** method draws a black rectangle with a top-left corner at position 20,20. The rectangle is 150 pixel wide and 100 pixels high.

Example

```
const myCanvas = document.getElementById("myCanvas");  
const ctx = myCanvas.getContext("2d");  
  
ctx.fillRect(20, 20, 150, 100);
```

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Using Colors

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Example

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```
const myCanvas = document.getElementById("myCanvas");
const ctx = myCanvas.getContext("2d");

ctx.fillStyle = "red";
ctx.fillRect(20, 20, 150, 100);
```

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You can also create a new `<canvas>` element with the `document.createElement()` method, and add the element to an existing HTML page:

Example

```
const myCanvas = document.createElement("canvas");
document.body.appendChild(myCanvas);
const ctx = myCanvas.getContext("2d");

ctx.fillStyle = "red";
ctx.fillRect(20, 20, 150, 100);
```

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Paths

The common way to draw on the canvas is to:

1. Begin a Path - `beginPath()`
2. Move to a Point - `moveTo()`
3. Draw in the Path - `lineTo()`
4. Draw the Path - `stroke()`

Example

```
const canvas = document.getElementById("myCanvas");
const ctx = canvas.getContext("2d");

ctx.beginPath();
ctx.moveTo(20, 20);
ctx.lineTo(20, 100);
ctx.lineTo(70, 100);
ctx.stroke();
```

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Complete Canvas API Reference

This reference covers all Properties and Methods of the `getContext("2d")` Object, used to draw text, lines, boxes, circles, pictures, and more on the canvas.

Drawing Methods

There are only 3 methods to draw directly on the canvas:

[strokeRect\(\)](#)
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Draws a rectangle (with no fill)

[clearRect\(\)](#)

Clears specified pixels within a rectangle

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Path Methods

Method	Description
beginPath()	Begins a new path or resets the current path
closePath()	Adds a line to the path from the current point to the start
isPointInPath()	Returns true if the specified point is in the current path
moveTo()	Moves the path to a point in the canvas (without drawing)
lineTo()	Adds a line to the the path
fill()	Fills the current path
rect()	Adds a rectangle to the path
stroke()	Draws the current path



arc() REMOVE ADS	Adds an arc/curve (circle, or parts of a circle) to the path
arcTo()	Adds an arc/curve between two tangents to the path
quadraticCurveTo()	Adds a quadratic Bézier curve to the path

Text

Method/Prop	Description
direction	Sets or returns the direction used to draw text
fillText()	Draws "filled" text on the canvas
font	Sets or returns the font properties for text content
measureText()	Returns an object that contains the width of the specified text
strokeText()	Draws text on the canvas
textAlign	Sets or returns the alignment for text content
textBaseline	Sets or returns the text baseline used when drawing text

Colors, Styles, and Shadows

Method/Property	Description
addColorStop()	Specifies the colors and stop positions in a gradient object
createLinearGradient()	Creates a linear gradient (to use on canvas content)
createPattern()	Repeats a specified element in the specified direction
createRadialGradient()	Creates a radial/circular gradient (to use on canvas content)



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~~lineCap~~~~Sets or returns the style of the end caps for a line~~[REMOVE ADS](#)
lineJoin

Sets or returns the type of corner created, when two lines meet

lineWidth

Sets or returns the current line width

miterLimit

Sets or returns the maximum miter length

shadowBlur

Sets or returns the blur level for shadows

shadowColor

Sets or returns the color to use for shadows

shadowOffsetX

Sets or returns the horizontal distance of the shadow from the shape

shadowOffsetY

Sets or returns the vertical distance of the shadow from the shape

strokeStyle

Sets or returns the color, gradient, or pattern used for strokes

Transformations

Method	Description
<u>scale()</u> .	Scales the current drawing bigger or smaller
<u>rotate()</u> .	Rotates the current drawing
<u>translate()</u> .	Remaps the (0,0) position on the canvas
<u>transform()</u> .	Replaces the current transformation matrix for the drawing
<u>setTransform()</u> .	Resets the current transform to the identity matrix. Then runs <u>transform()</u> .

Image Drawing

Method	Description
<u>drawImage()</u> .	Draws an image, canvas, or video onto the canvas



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Method/Property Description

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[createImageData\(\)](#). Creates a new, blank ImageData object

[getImageData\(\)](#). Returns an ImageData object that copies the pixel data for the specified rectangle on a canvas

[ImageData.data](#) Returns an object that contains image data of a specified ImageData object

[ImageData.height](#) Returns the height of an ImageData object

[ImageData.width](#) Returns the width of an ImageData object

[putImageData\(\)](#). Puts the image data (from a specified ImageData object) back onto the canvas

Compositing

Property	Description
<u>globalAlpha</u>	Sets or returns the current alpha or transparency value of the drawing
<u>globalCompositeOperation</u>	Sets or returns how a new image are drawn onto an existing image

Other Methods

Method	Description
<u>clip()</u> .	Clips a region of any shape and size from the original canvas
<u>save()</u> .	Saves the state of the current drawing context and all its attributes
<u>restore()</u> .	Restores the previously saved state and attributes
<u>createEvent()</u>	

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Standard Properties and Events

The canvas object also supports the standard [properties](#) and [events](#).

Related Pages

Canvas tutorial: [Canvas Tutorial](#)

HTML tutorial: [HTML5 Canvas](#)

HTML reference: [HTML <canvas> tag](#)

Browser Support

The **<canvas>** element is an HTML5 standard (2014).

Canvas API is supported in all modern browsers:

Chrome	Edge	Firefox	Safari	Opera	IE
Yes	Yes	Yes	Yes	Yes	9-11

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